School Curriculum and Standards Authority

JUDGING STANDARDS

Reporting against the Achievement Standard

The annotated work samples in *Judging Standards* support teachers when reporting against the achievement standards; when giving assessment feedback; and when explaining the differences between one student's achievement and another's.

Grey highlighting identifies those aspects of the achievement standard addressed in the work sample. Annotations in black text refer to the assessment pointers while those in coloured text highlight additional, specific qualities evident in the work.

WORK SAMPLE: YEAR 10 SCIENCE

PERFORMANCE ASSOCIATED WITH GRADE C, REPRESENTING SATISFACTORY ACHIEVEMENT

SUMMARY OF TASK:

Gene technology: Genetically modified corn

Students selected an area of gene technology to research and presented their information on a poster using text and graphics. Part of the task involved identifying positives and negatives related to this type of technology.

YEAR 10 SCIENCE ACHIEVEMENT STANDARD

By the end of Year 10, students analyse how the periodic table organises elements and use it to make predictions about the properties of elements. They explain how chemical reactions are used to produce particular products and how different factors influence the rate of reactions. They explain the concept of energy conservation and represent energy transfer and transformation within systems. They apply relationships between force, mass and acceleration to predict changes in the motion of objects. Students describe and analyse interactions and cycles within and between Earth's spheres. They evaluate the evidence for scientific theories that explain the origin of the universe and the diversity of life on Earth. They explain the processes that underpin heredity and evolution. Students analyse how the models and theories they use have developed over time and discuss the factors that prompted their review.

Students develop questions and hypotheses and independently design and improve appropriate methods of investigation, including field work and laboratory experimentation. They explain how they have considered reliability, safety, fairness and ethical actions in their methods and identify where digital technologies can be used to enhance the quality of data. When analysing data, selecting evidence and developing and justifying conclusions, they identify alternative explanations for findings and explain any sources of uncertainty. Students evaluate the validity and reliability of claims made in secondary sources with reference to currently held scientific views, the quality of the methodology and the evidence cited. They construct evidence-based arguments and select appropriate representations and text types to communicate science ideas for specific purposes.

KEY WORDS:

Genes, DNA, Genetic manipulation, Genetic engineering

Student achievement is reported at the end of the semester or year using the letter grades and achievement descriptors. Letter grades and achievement descriptors should not be used to assess individual pieces of work.

WORK SAMPLE

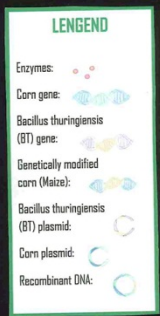
GENETICALLY MODIFIED CORN

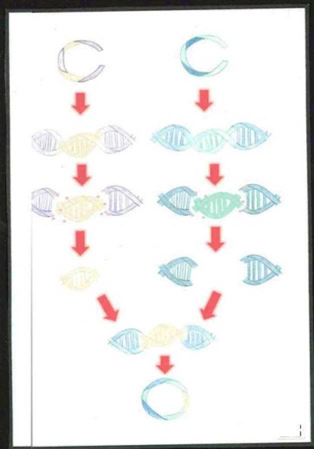
[redacted]

POSITIVES	NEGATIVES
Reduced demand for pest Resistance: Farmers usually spend thousands of dollars on chemical pesticides each year. However, growing genetically modified foods such as BT Corn can help reduce the annual cost of pesticides and eliminate the number of risks that evolve due to these chemicals. Without GM corn, we run the possibility of causing potential health hazards, harm to the environment and poisoning water supply all due to the pesticide spray that farmers use on their corn crops. Therefore GM corn saves the farmers money and helps to protect our environment and people.	**Monarch Butterfly:** GM corn can cause unintended harm to many organisms such as the Monarch Butterfly. The butterflies are at risk of death during their caterpillar stage. Monarch caterpillar's diet consists of milkweed plants, not corn, however it is becoming a concerning problem that winds are cross pollinating neighbouring BT corn and milkweed plant crops. This is creating toxic pollen which is covering the milkweed plants causing the caterpillars to eat this and die. With this ongoing problem, it is likely that the Monarch butterfly could soon become extinct.
Food supply: Genetically modifying corn can allow farmers to harvest more produce at a cheaper price. This in turn, lowers the corn prices for consumers. GM corn develops at a faster rate as it is able to grow and thrive in environments that may have been unavailable due to insects before.	**Cancer:** A study in late 2012, displayed that GM corn causes cancer in rats. The study found that tumours in rats of both sexes fed the GM corn were two to three times larger than in the control group (thewest.com.au, 2012). This raised many question about the safety of GM corn. An investigation was ordered by the French government however, so far there has been no strong links to cancer in humans from GM corn.
Nutrients: We can increase the amounts of vitamins and many other nutrients that are available to us when producing genetically modified corn. This ability is predominantly beneficial in parts of the world where nutritional scarcity is common.	**Allergies:** It is possible that genetically modified corn could produce allergies to animals and humans. BT corn has been linked to allergies in humans however it has not caused a common effect yet.
Better produce: Genetically modified corn repeals insects from damaging the crops through the use of chemical herbicides that have been inserted into the corns DNA. This allows the corn to grow without the disturbance of insects harming the crop, therefore producing better quality corn.	**Change to crop diversity:** Modifying the genetic sections and construction of corn DNA can change the crop diversity. Altering corn DNA can cause the taste, colour, growing conditions, size, height, fruiting time or even cooking conditions to vary, resulting in a different crop. If we continue to modify corn, we could in fact transform it into a very different looking and tasting plant which may come with the 'extinction' of corn.

ADDITIONAL POINTS

- 86% of corn is genetically modified in the US.
- By 2012, 26 assortments of herbicide-resistant GM maize were approved for import into the European Union.
- At this present time, there aren't any known human health risks associated with Bt corn.
- Genetically modified food has been sold commercially since 1994.
- Herbicide-resistant GM corn is grown in 14 countries around the globe.





HOW DNA IS ALTERED

DNA is altered through the use of genetic engineering which is also known as genetic modification. The process involves the direct modification of an organism's characteristics by manipulating its genetic material using biotechnology. With the use of particular techniques and modern technology, we can alter the genetic makeup of and organism. This involves using DNA techniques to form new combinations of genetic material to either be inserted it into the organism directly or indirectly.

New DNA can be injected into the host gnome directly. This is done by firstly isolating and then copying the required material using cloning techniques to produce the necessary DNA sequence. DNA can also be injected into the gnome indirectly. This is done by manufacturing the DNA and then inserting it into the required organism.

An organism that undergoes any form of DNA alteration is considered to be a genetically modified organism (GMO). The first modified organism was bacteria in 1973. Since then many other organism have been modified and sold commercially across the globe such as this topic, GM corn.

THE PROCESS OF GENETICALLY MODIFYING CORN

Genetically modifying corn is a multistep process. It involves gene splicing which requires the use of enzymes to cut open plasmids and insert the desired gene. The technology of merging DNA from different genes is called recombinant DNA. Both these steps are shown in the diagram. Firstly scientists isolate a Bacillus thuringiensis gene. Scientists then cut a plasmid with the use of enzymes that have been extracted from bacterium of corn. Next they insert the Bacillus thuringiensis gene into the corn plasmid to form recombinant DNA. Scientists then put the recombinant DNA back into the bacterium of corn. The bacterial cells grow and divide to produce many copies of the new gene. The original corn has now been transformed into genetically modified corn also known as Bt corn. Bacillus thuringiensis was inserted into corn as it is a soil bacterium that is commonly used as a biological pesticide which produces a protein that exterminates insects.

ANNOTATIONS

Presents scientific ideas and information using appropriate headings, including some representations.

WORK SAMPLE

POSITIVES

Reduced demand for pest Resistance:

Farmers usually spend thousands of dollars on chemical pesticides each year. However, growing genetically modified foods such as BT Corn can help reduce the annual cost of pesticides and eliminate the number of risks that evolve due to these chemicals. Without GM corn, we run the possibility of causing potential health hazards, harm to the environment and poisoning water supply all due to the pesticide spray that farmers use on their corn crops. Therefore GM corn saves the farmers money and helps to protect our environment and people.

Food supply:

Genetically modifying corn can allow farmers to harvest more produce at a cheaper price. This in turn, lowers the corn prices for consumers. GM corn develops at a faster rate as it is able to grow and thrive in environments that may have been unavailable due to insects before.

Nutrients:

We can increase the amounts of vitamins and many other nutrients that are available to us when producing genetically modified corn. This ability is predominantly beneficial in parts of the world where nutritional scarcity is common.

Better produce:

Genetically modified corn repeals insects from damaging the crops through the use of chemical herbicides that have been inserted into the corns DNA. This allows the corn to grow without the disturbance of insects harming the crop, therefore producing better quality corn.

NEGATIVES

Monarch Butterfly:

GM corn can cause unintended harm to many organisms such as the Monarch Butterfly. The butterflies are at risk of death during their caterpillar stage. Monarch caterpillar's diet consists of milkweed plants, not corn, however it is becoming a concerning problem that winds are cross pollinating neighbouring BT corn and milkweed plant crops. This is creating toxic pollen which is covering the milkweed plants causing the caterpillars to eat this and die. With this ongoing problem, it is likely that the Monarch butterfly could soon become extinct.

Cancer:

A study in late 2012, displayed that GM corn causes cancer in rats. The study found that tumours in rats of both sexes fed the GM corn were two to three times larger than in the control group (thewest.com.au, 2012). This raised many question about the safety of GM corn. An investigation was ordered by the French government however, so far there has been no strong links to cancer in humans from GM corn.

Allergies:

It is possible that genetically modified corn could produce allergies to animals and humans. BT corn has been linked to allergies in humans however it has not caused a common effect yet.

Change to crop diversity:

Modifying the genetic sections and construction of corn DNA can change the crop diversity. Altering corn DNA can cause the taste, colour, growing conditions, size, height, fruiting time or even cooking conditions to vary, resulting in a different crop. If we continue to modify corn, we could in fact transform it into a very different looking and tasting plant which may come with the 'extinction' of corn.

ADDITIONAL POINTS

- 86% of corn is genetically modified in the US.
- By 2012, 26 assortments of herbicide-resistant GM maize were approved for import into the European Union.
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- Genetically modified food has been sold commercially since 1994.
- Herbicide-resistant GM corn is grown in 14 countries around the globe.

ANNOTATIONS

Explains changes to a system using basic understanding of a science theory, e.g. life cycle of the Monarch butterfly; possible environmental harm and risk to humans as a result of farmers' use of pesticides.

Explains scientific concepts in simple terms. Makes occasional errors or unsupported claims, e.g. makes a generalisation about nutrients which is not related to pesticide resistance.

Confuses herbicides and pesticides: 'Genetically modified corn repeals (sic) insects from damaging the cops through the use of chemical herbicides…'

WORK SAMPLE | **ANNOTATIONS**

HOW DNA IS ALTERED

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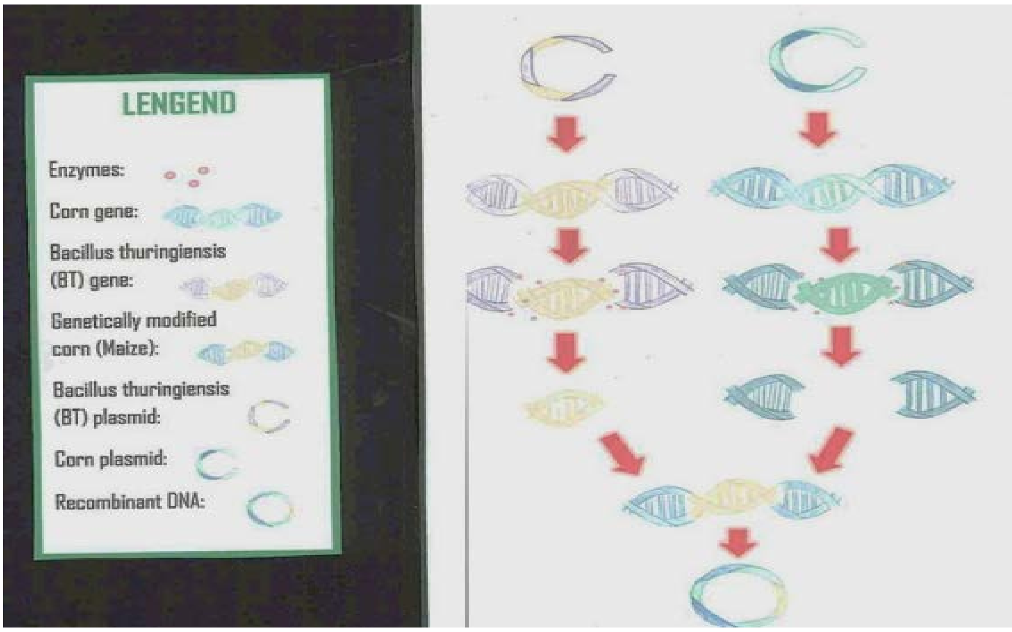
An organism that undergoes any form of DNA alteration is considered to be a genetically modified organism (GMO). The first modified organism was bacteria in 1973. Since then many other organism have been modified and sold commercially across the globe such as this topic, GM corn.

Outlines scientific concepts and processes, using correct scientific terminology and knowledge, e.g. describes some aspects of how DNA is altered. (*Some explanations such as this one show performance qualities of a B [High] standard.)

THE PROCESS OF GENETICALLY MODIFYING CORN

Genetically modifying corn is a multistep process. It involves gene splicing which requires the use of enzymes to cut open plasmids and insert the desired gene. The technology of merging DNA from different genes is called recombinant DNA. Both these steps are shown in the diagram. Firstly scientists isolate a Bacillus thuringiensis gene. Scientists then cut a plasmid with the use of enzymes that have been extracted from bacterium of corn. Next they insert the Bacillus thuringiensis gene into the corn plasmid to form recombinant DNA. Scientists then put the recombinant DNA back into the bacterium of corn. The bacterial cells grow and divide to produce many copies of the new gene. The original corn has now been transformed into genetically modified corn also known as Bt corn. Bacillus thuringiensis was inserted into corn as it is a soil bacterium that is commonly used as a biological pesticide which produces a protein that exterminates insects.

Explains scientific concepts in simple terms, using some scientific terminology in the correct context, e.g. explains the process of genetically modifying corn but with some errors and omissions.



Uses diagrams to support explanations of scientific models or processes; however, diagrams contains occasional errors and/or lack some information, e.g. uses a simple diagram with a legend to illustrate the process of genetically modifying corn, but the absence of labels for parts of the process make the representation unclear.